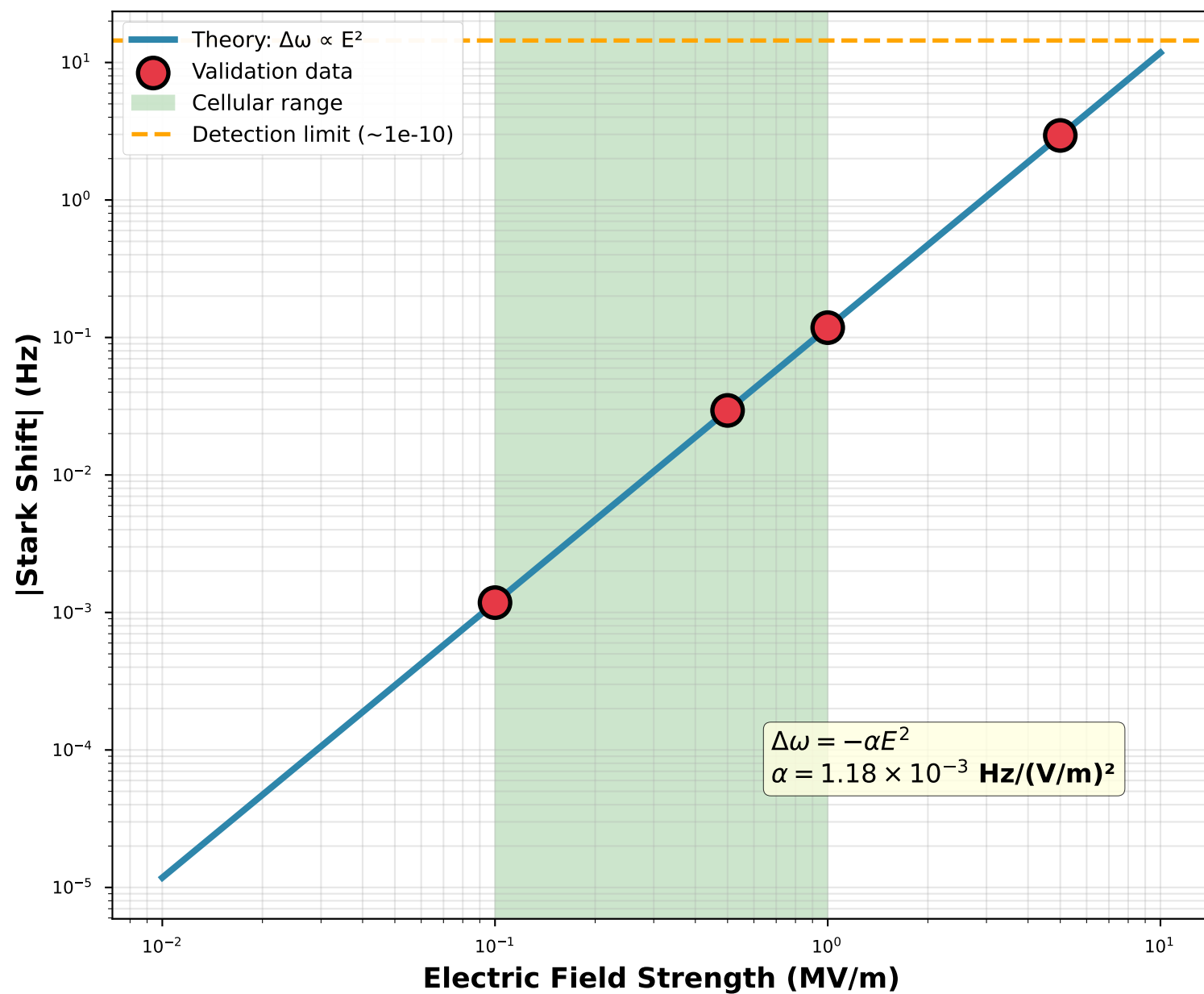
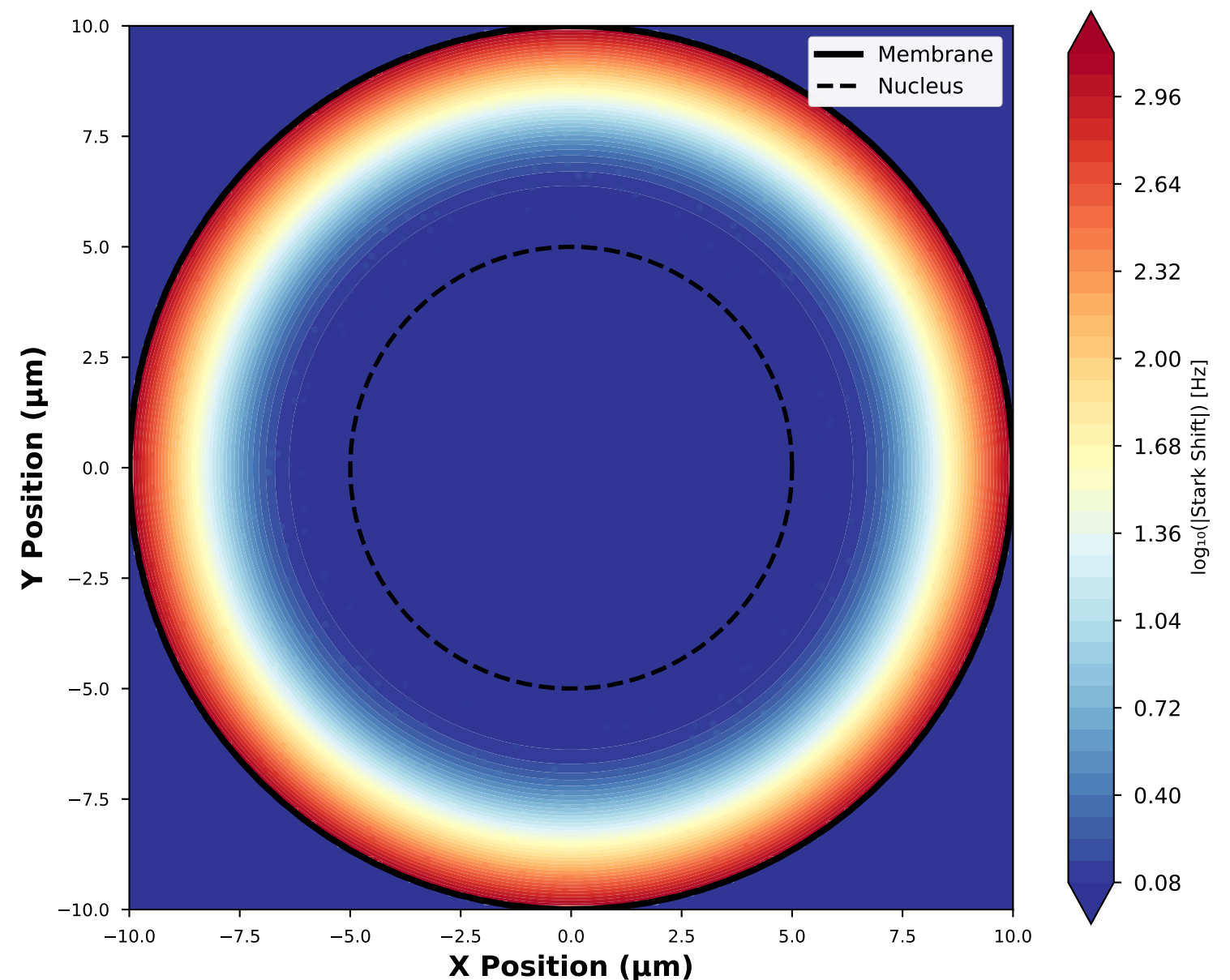


Figure 3: O₂ Stark Spectroscopy for Cellular Field Measurement

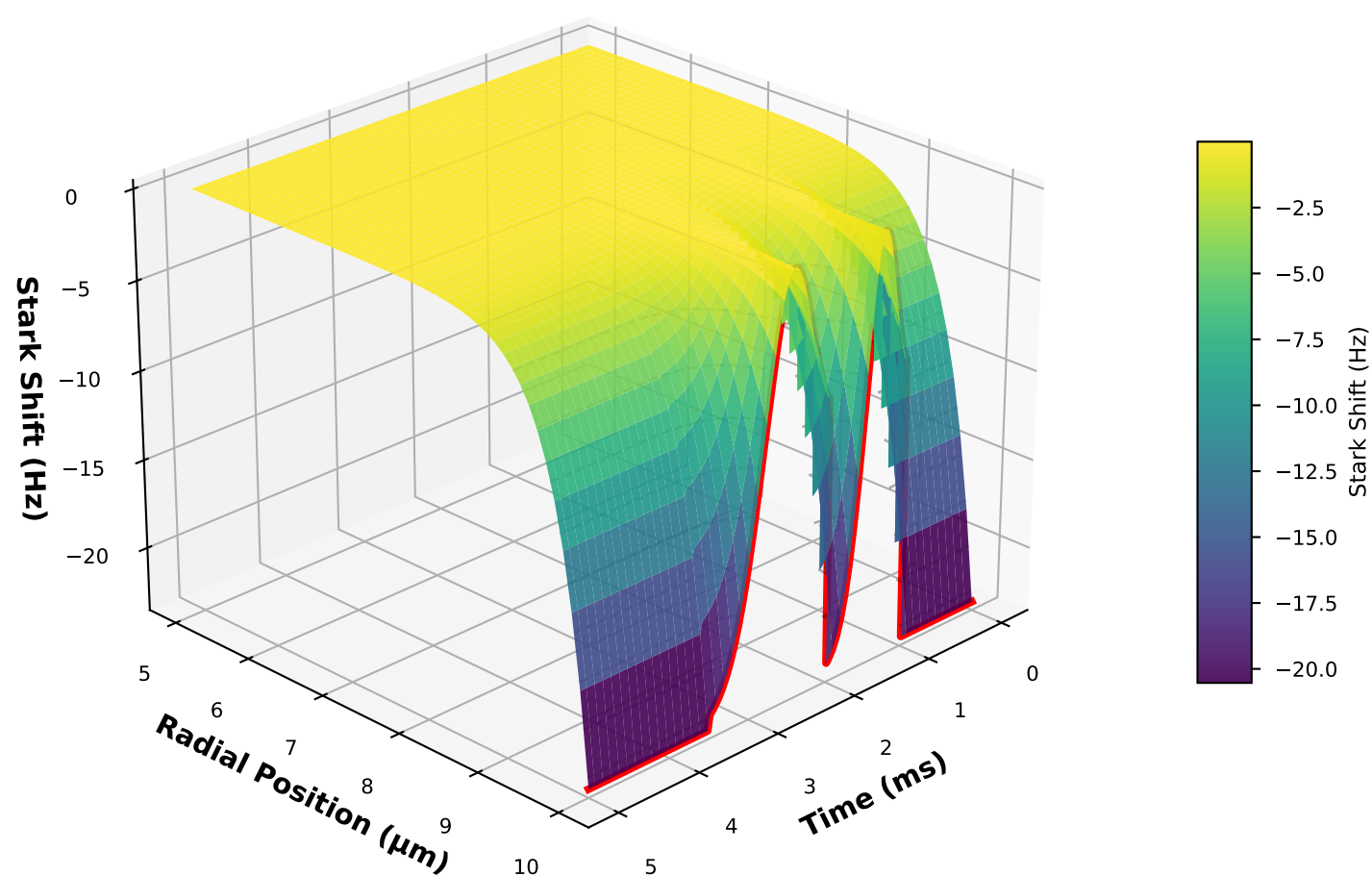
(A) O₂ Stark Shift: Quadratic Field Dependence



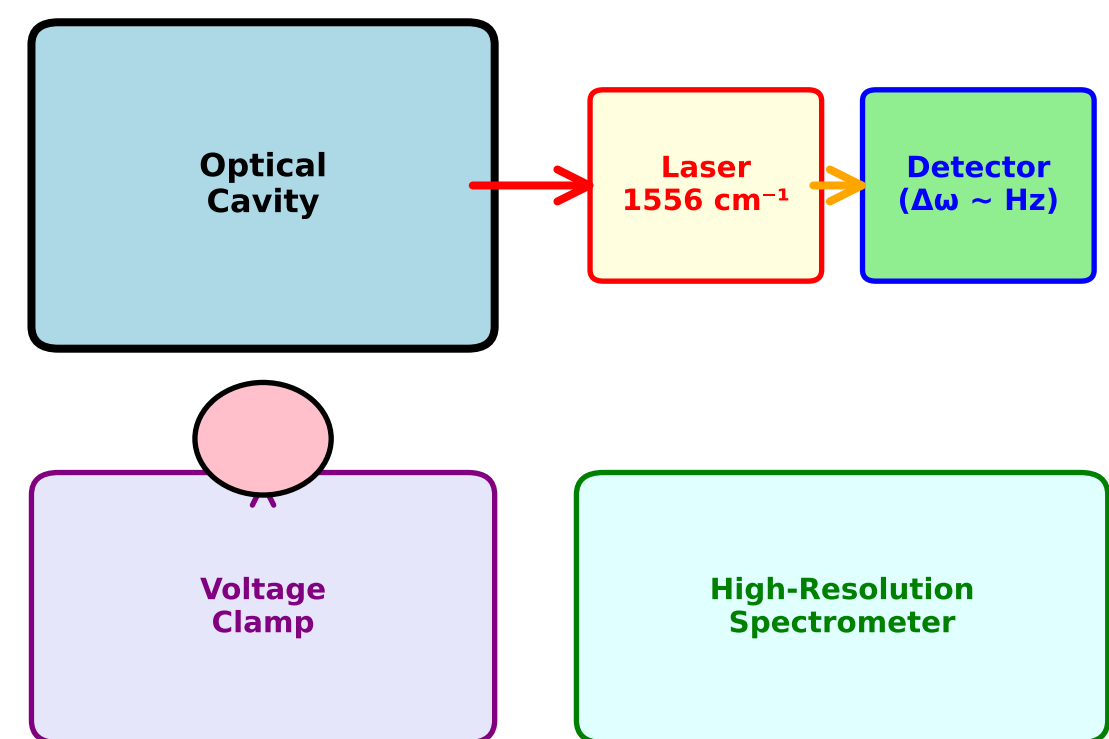
(B) Spatial Stark Shift Map: O₂ Sensor Array



(C) Temporal Dynamics: Action Potential Modulation



(D) Proposed Experimental Setup: Cavity-Enhanced Stark Spectroscopy



Technical Specifications:

- Frequency: O₂ ν_1 band (1556 cm^{-1})
- Resolution: 0.001 cm^{-1} (30 MHz)
- Sensitivity: 10^{-10} relative shift
 - Field range: 0.1-10 MV/m
 - Temperature: 310 K (37°C)
- Method: Cavity-enhanced Raman